Mathematical Modeling

Chapter 2: Linear Programming—Models and Geometry

Personal study notes

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1 The linear programming problem

A linear program (LP, Polish abbreviations PL or ZPL) optimizes a linear function $c^T x$ subject to linear equality and inequality constraints, with continuous variables. The feasible set Ω includes every sign and bound restriction. For maximization, a feasible x^* is globally optimal when $c^T x^* \geq c^T x$ for every $x \in \Omega$.

Minimization and maximization are related by

$$\min_{x\in\Omega} c^T x = -\max_{x\in\Omega}(-c)^T x.$$

The optimizers agree after this sign change; the objective values have opposite signs. An equivalent reformulation with additional variables requires a mapping back to original decisions, not literal equality of vector sets in different dimensions. Model construction follows Lecture 1: variables, constraints, objective, algorithm, verification, and post-optimal analysis.

2 Production planning: formulation and full hand solution

Two products consume three resources. The data are:

	Product 1	Product 2	Available
Resource 1 (kg/item; availability in kg)	2	1	10
Resource 2 (kg/item; availability in kg)	3	3	24
Resource 3 (kg/item; availability in kg)	1.5	0	6
Selling price (PLN/item)	3	2	—

Let x_1, x_2 be production quantities. Assume constant coefficients, additive resource usage, and fractional production allowed. Since costs are absent, $3x_1 + 2x_2$ is *revenue*. Computing profit would also require costs. The model is

$$\max(3x_1 + 2x_2), \qquad \begin{cases} 2x_1 + x_2 \le 10, \\ 3x_1 + 3x_2 \le 24, \\ 1.5x_1 \le 6, \\ x_1, x_2 \ge 0. \end{cases} \tag{1}$$

The second inequality is $x_1 + x_2 \le 8$, and the third is $x_1 \le 4$. Intersect boundary lines and retain only feasible intersections:

Vertex	x_1	x_2	$3x_1 + 2x_2$
V_1	0	0	0
V_2	4	0	12
V_3	4	2	16
V_4	2	6	18
V_5	0	8	16

For V_4, subtract $x_1 + x_2 = 8$ from $2x_1 + x_2 = 10$ to obtain $x_1 = 2$, then $x_2 = 6$. These give resource slacks $(0, 0, 3)^T$ kg.

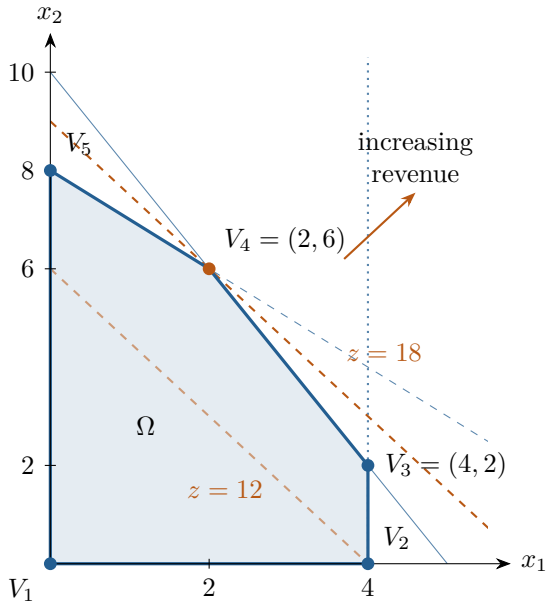


Figure 1: Production constraints bound the shaded feasible polygon. The solid, dashed, and dotted resource boundaries are $2x_1 + x_2 = 10$, $x_1 + x_2 = 8$, and $x_1 = 4$. Orange lines are levels of $z = 3x_1 + 2x_2$; the last feasible contact is V_4, where the first two resources are exhausted.

The objective level lines have slope $-3/2$. Moving them toward increasing revenue brings the last feasible contact to $(2, 6)$.

2.1 An optimality certificate without a graph

For every feasible point,

$$3x_1 + 2x_2 = (2x_1 + x_2) + (x_1 + x_2) \leq 10 + 8 = 18.$$

The candidate $(2, 6)$ attains the bound, proving global optimality. Equality requires both resource inequalities to be tight, whose unique intersection is $(2, 6)$; thus this optimum is unique. This reasoning is more informative than merely listing a solver output.

3 Other modeling patterns

3.1 Diet problem

Let x_j be the daily quantity of food j, c_j its unit price, and a_{ij} the amount of nutrient i per food unit. Nutrient limits are b_i, d_i, while food-quantity bounds are g_j, h_j. The model is

$$\min \sum_{j=1}^{n} c_j x_j, \qquad b_i \leq \sum_{j=1}^{n} a_{ij} x_j \leq d_i \ (i = 1, \ldots, m), \quad g_j \leq x_j \leq h_j, \quad x_j \geq 0.$$

Nutrient and food bounds have different units and meanings. Positive food lower bounds can enforce minimum variety, but generic bounds do not automatically produce a realistic diet.

For a hand-checkable instance, let x and y be the daily numbers of servings of foods 1 and 2, respectively. Fractional servings are allowed: $x, y \geq 0$. The following table gives the price of one serving of each food and the amounts of two nutrients it supplies. At least 4 g of each nutrient is required per day.

Per serving	Food 1	Food 2
Price (PLN)	2	3
Nutrient 1 (g)	2	1
Nutrient 2 (g)	1	3

For example, one serving of food 2 supplies 1 g of nutrient 1 and 3 g of nutrient 2. Thus x servings of food 1 and y servings of food 2 supply $2x + y$ grams of nutrient 1 and $x + 3y$ grams of nutrient 2. The resulting model is

$$\min_{x,y \geq 0} 2x + 3y, \qquad 2x + y \geq 4, \quad x + 3y \geq 4.$$

To find a candidate, set both nutrient requirements to equality. Substituting $y = 4 - 2x$ into $x + 3y = 4$ gives $x = 8/5$ and $y = 4/5$. These nonnegative quantities meet both requirements and cost 28/5 PLN per day. An intersection alone does not prove optimality. Weighting the two requirements by 3/5 and 4/5 gives, for every feasible pair,

$$2x + 3y = \tfrac{3}{5}(2x + y) + \tfrac{4}{5}(x + 3y) \geq \tfrac{28}{5}.$$

The candidate reaches this lower bound, proving optimality. Both weights are positive, so attaining the bound requires both nutrient constraints to be tight; their unique intersection also proves uniqueness. The inequality directions express minimum intake; changing them to upper bounds would describe a different requirement. Upper nutrient and serving bounds, when specified, must also be checked before accepting a candidate.

3.2 Transportation problem

Let x_{ij} be the quantity shipped from supplier i to customer j; a_i is available supply, b_j required demand, and c_{ij} cost per shipped unit. A formulation with supply limits and minimum demand is

$$\min \sum_{i=1}^{m} \sum_{j=1}^{n} c_{ij} x_{ij}, \qquad \sum_{j=1}^{n} x_{ij} \leq a_i, \quad \sum_{i=1}^{m} x_{ij} \geq b_j, \quad x_{ij} \geq 0.$$

With nonnegative supply/demand and every route available without capacity limits, feasibility is equivalent to $\sum_i a_i \geq \sum_j b_j$. Necessity follows by summing constraints; sufficiency follows by allocating supplies successively to remaining demands. With restricted routes, the total-supply test alone is insufficient.

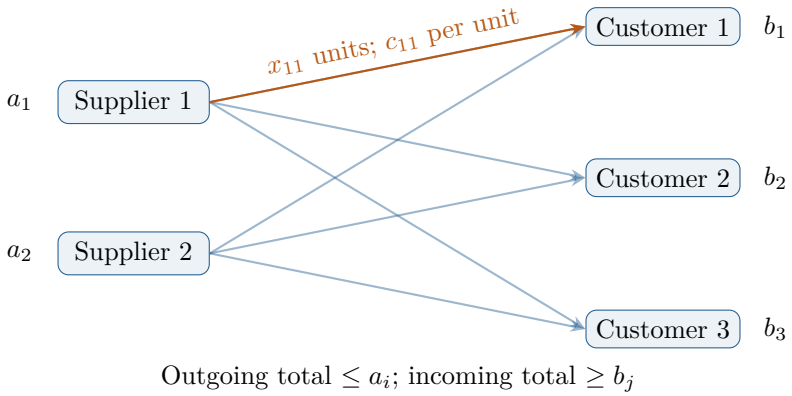


Figure 2: A two-supplier, three-customer transportation network. Each arrow represents a decision x_{ij}; supply and demand constraints sum the flows at the corresponding node. The displayed network allows every route.

If all supply must be shipped and every demand met exactly, use equalities and balanced totals instead. These are different modeling requirements.

3.3 Regression as a linear program

For observed pairs (t_i, y_i), fit $y \approx at + b$, with residuals $e_i = y_i - at_i - b$. The t_i are given observations, not decision variables. The slope a and intercept b are free real variables. Minimizing the largest absolute residual introduces $r \geq 0$:

$$\begin{aligned} \min_{a,b,r} \quad & r, \\ at_i + b - r &\leq y_i, \\ -at_i - b - r &\leq -y_i \qquad (i = 1, \ldots, N). \end{aligned}$$

Indeed, the two inequalities are equivalent to $-r \leq e_i \leq r$. For the alternative least-absolute-deviation criterion, introduce one $r_i \geq 0$ per observation and minimize $\sum_i r_i$ under $-r_i \leq e_i \leq r_i$. Least squares minimizes $\sum_i e_i^2$, a quadratic objective rather than a linear one. The three error criteria need not produce the same fitted line.

To see this difference by hand, use observations $(0, 0), (1, 2), (2, 0)$, indexed here by $i = 0, 1, 2$. For every affine line, the residuals satisfy

$$e_1 - \frac{e_0 + e_2}{2} = 2.$$

(Substitute $e_i = y_i - at_i - b$ for the three observations; the a and b terms cancel.) If $|e_i| \leq r$, the left side is at most $2r$, so $r \geq 1$. The constant line $a = 0, b = 1$ attains $r = 1$, establishing the minimax optimum. The same identity gives $2 \leq |e_1| + (|e_0| + |e_2|)/2 \leq \sum_i |e_i|$. The constant line $a = b = 0$ attains total absolute error 2 and is therefore optimal for that criterion. For fixed a, b, choose the shared minimax bound $r = \max_i |e_i|$; for the sum criterion, choose each $r_i = |e_i|$. Enlarging these bounds cannot improve either minimization objective. The formulas are LPs because observation coordinates are fixed and only a, b, r or r_i are decision variables.

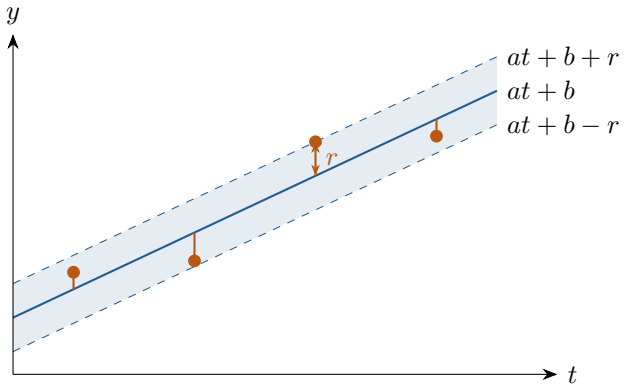


Figure 3: The minimax regression constraints place all observations inside a vertical error band. Here the illustrative line is $y = t + 1$ and $r = 0.6$; one observation reaches the upper boundary. Residuals measure vertical distance at fixed t_i, rather than perpendicular distance to the line.

4 Cutting boards and integer decisions

We buy long boards and cut them into the shorter pieces required for cabinets. The length L is the full length of one board before cutting. A *cutting pattern* specifies which pieces to cut from a single board; the unused remainder is its *offcut*.

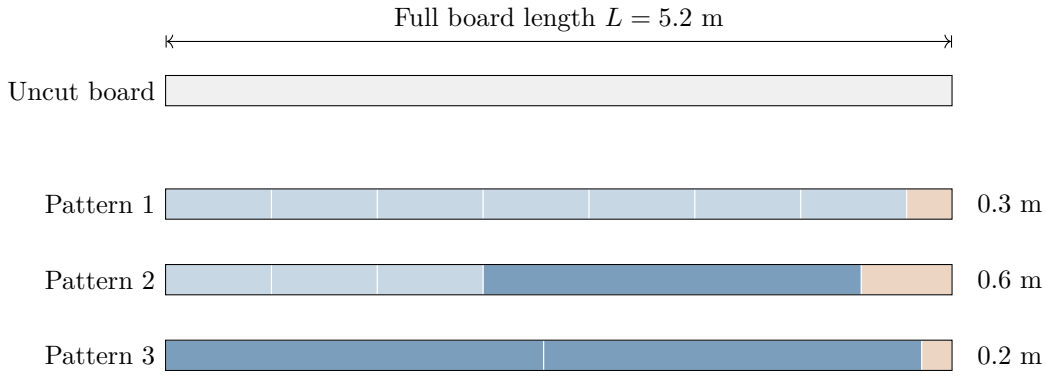


Figure 4: Each pattern cuts one full board into shorter pieces. For example, pattern 2 gives three 0.7 m pieces and one 2.5 m piece, leaving $5.2 - 3(0.7) - 2.5 = 0.6$ m unused.

Let ℓ_i be the required piece lengths and b_i their required counts. Pattern j produces a_{ij} pieces of length ℓ_i per board, with nonnegative integer a_{ij} and $\sum_i a_{ij}\ell_i \le L$. Its offcut is

$$r_j = L - \sum_i a_{ij}\ell_i.$$

Let x_j count how many boards are cut using pattern j, and let $A = (a_{ij})$. For exact demand, minimize $r^T x$ subject to $Ax = b$, $x \ge 0$. For minimum demand, replace $Ax = b$ by $Ax \ge b$, permitting overproduction. Physical counts require integer x_j; real-valued x_j define an LP relaxation.

The condition $r_j < \min_i \ell_i$ describes *maximal* patterns: no additional piece fits. It is not a requirement for every feasible pattern. Under exact demand, nonmaximal patterns may be needed to avoid excess pieces.

4.1 Cabinet example with 5.2 m boards: full hand solution

At least 200 cabinet sets require 1400 short pieces (0.7 m) and 800 long pieces (2.5 m). The three permitted cutting patterns are:

	Pattern 1	Pattern 2	Pattern 3
Short pieces	7	3	0
Long pieces	0	1	2
Offcut (m)	0.3	0.6	0.2

We optimize over these supplied patterns; they are not an enumeration of all length-feasible cuts. For instance, six short pieces and no long ones would be a feasible but nonmaximal cut. Assume the listed patterns are the permitted ones, ignore material lost to the saw (kerf), and allow surplus pieces:

$$\min 0.3x_1 + 0.6x_2 + 0.2x_3, \qquad 7x_1 + 3x_2 \geq 1400, \quad x_2 + 2x_3 \geq 800, \quad x \geq 0.$$

The candidate $(200, 0, 400)^T$ meets both demands exactly and has offcut $60 + 80 = 140$ m. To prove optimality, take nonnegative multiples $3/70$ and $1/10$ of the two demand inequalities. Their left-hand side is

$$\frac{3}{10}x_1 + \frac{8}{35}x_2 + \frac{1}{5}x_3 \geq 140.$$

Since $0.6 - 8/35 = 13/35 > 0$,

$$0.3x_1 + 0.6x_2 + 0.2x_3 \geq 140 + \frac{13}{35}x_2 \geq 140.$$

Equality requires $x_2 = 0$ and both demands tight, yielding the stated candidate. It is the unique optimum for the supplied continuous model and, being integer, also solves its integer version. It uses 600 boards.

Remark 4.1 (Offcut versus number of boards used)**.** With exact demand, $r^T x = L\sum_j x_j - \sum_i \ell_i b_i$, so minimizing offcut and minimizing the number of boards are equivalent. With minimum demand, produced length can vary; the objectives are not generally equivalent. Account separately for unwanted surplus, saw kerf, and limits on available boards when the real task requires them.

4.2 Seven cabinets from 2 m boards

The original prompt is:

> Ile desek o długości d=2 m potrzeba na wykonanie 7 szafek?

English translation: How many 2 m boards are needed to make seven cabinets? The sketch shows three horizontal members and two vertical members. Interpret the marked dimensions as cut lengths: three 0.8 m and two 0.3 m pieces per cabinet. This assumes zero kerf and ignores board thickness and joint allowances; the sketch does not specify those details. It gives demands 21 long and 14 short pieces.

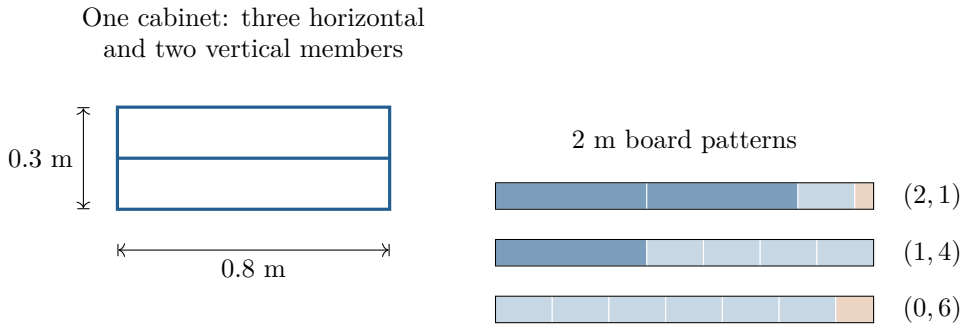


Figure 5: The cabinet interpretation and its maximal board patterns. Lengths refer to the stated idealized cut lengths, with zero kerf and no joint allowances. The plan $10(2,1)+1(1,4) = (21,14)$ supplies seven cabinets.

Writing patterns as (long pieces, short pieces), the maximal patterns are $(2,1)$, $(1,4)$, and $(0,6)$. Let their counts be $u, v, w \in \{0,1,2,\ldots\}$. The model is

$$\min u + v + w, \qquad 2u + v \geq 21, \quad u + 4v + 6w \geq 14.$$

Required total length is $21(0.8) + 14(0.3) = 21$ m, so at least $\lceil 21/2 \rceil = 11$ boards are necessary. Ten $(2,1)$ boards and one $(1,4)$ board produce exactly $(21,14)$ pieces, using 11 boards and leaving 1 m of offcut. This construction attains the length lower bound and proves optimality under the stated interpretation. It is not a claim about unspecified manufacturing details.

4.3 What an integer relaxation can and cannot certify

Removing integrality enlarges the feasible set. Thus, for maximization, $z^*_{\mathrm{IP}} \leq z^*_{\mathrm{LP}}$ whenever both optima exist; for minimization the bound reverses. Any integer feasible candidate supplies the opposite bound. An integer LP optimizer therefore certifies an integer optimum, as in the 5.2 m cutting example. An infeasible relaxation proves the integer model infeasible; an unbounded relaxation alone need not prove the integer model unbounded, since it may have no integer feasible point.

Consider the small integer model

$$\max \; 2u + 3v, \qquad u + 2v \leq 4, \quad 0 \leq u \leq 5/2, \quad v \geq 0, \quad u, v \text{ integer}.$$

In its LP relaxation, $2u+3v = \frac{3}{2}(u+2v)+\frac{1}{2}u \leq 6+5/4 = 29/4$. The point $(5/2, 3/4)$ attains this bound. Rounding both coordinates down gives $(2,0)$, of value 4, whereas the integer optimum $(2,1)$ has value 7. To prove the latter, integrality gives $u \leq 2$, hence $2u + 3v \leq 6 + u/2 \leq 7$, attained at $(2,1)$. Rounding to the nearest integers gives $v = 1$ and $u = 2$ or 3, depending on the tie convention; choosing 3 violates both upper constraints.

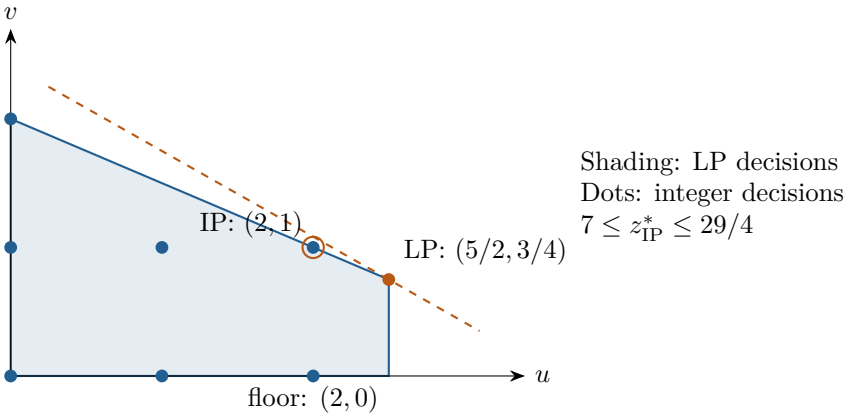


Figure 6: A relaxation gives a bound; rounding gives a candidate that still needs feasibility and optimality checks. Here flooring loses value 3.

Branch and bound turns relaxation bounds into an exact search. At a fractional coordinate u^*, every integer solution lies in one of $u \le \lfloor u^* \rfloor$ and $u \ge \lceil u^* \rceil$. Solve the child relaxations. If a child has an integer optimizer, update the best known integer solution (the *incumbent*) when it improves the objective, then close that child. Also close an infeasible child or one whose upper bound cannot exceed the incumbent value. In this example the branches are $u \le 2$ and $u \ge 3$: the second is infeasible, and the first has the integer LP optimum $(2, 1)$ with bound 7. This closes the search. In larger problems, cuts can tighten the bounds by excluding fractional points while preserving all integer feasible points. Ordinary simplex alone does not enforce integrality.

5 Standard form and equivalent transformations

We use the convention

$$\max c^T x, \qquad Ax = b, \quad x \ge 0, \quad A \in \mathbb{R}^{m \times n}, \quad b \in \mathbb{R}^m, \quad c \in \mathbb{R}^n. \tag{2}$$

Other texts use different "standard form" conventions. Here it means equality constraints and nonnegative variables. Full row rank is a useful basis assumption, not part of the definition of every LP. Remove redundant equalities after checking consistency; $m < n$ is a common case, not a universal requirement.

Original condition	Replacement
$a^T x \le b$	$a^T x + s = b$, $s \ge 0$ (slack)
$a^T x \ge b$	$a^T x - s = b$, $s \ge 0$ (surplus)
$\ell \le a^T x \le u$	Two inequalities, transformed separately
$x_j \in \mathbb{R}$	$x_j = x_j^+ - x_j^-$, $x_j^+, x_j^- \ge 0$
$x_j \le 0$	$x_j = -u_j$, $u_j \ge 0$
$x_j \ge \ell_j$	$x_j = \ell_j + y_j$, $y_j \ge 0$

A finite upper bound becomes $y_j \le u_j - \ell_j$ after the last substitution. Substitution may add a constant to the objective; retain it when reporting the original objective value. Free-variable splitting is not unique: adding the same positive number to x_j^+ and x_j^- leaves x_j unchanged.

Each slack or surplus has a meaning: $s = b - a^T x$ for an upper bound and $s = a^T x - b$ for a lower bound. Its nonnegativity is exactly the original inequality. Conversely, a nonnegative slack satisfying the equality recovers that inequality. When multiplying an inequality by -1, reverse its direction; when multiplying an equality by -1, retain equality. Reversing a minimization objective to maximize its negative preserves the optimizers and reverses the optimal value's sign.

These operations require recovering the original objective and variable domains before reporting an answer.

5.1 A four-variable conversion example

The original problem is

$$\min 2x_1 - x_2 + 5x_3 - 2x_4, \qquad \begin{cases} x_1 + x_2 + 2x_3 + 5x_4 \le 20, \\ 2x_1 - x_2 - x_4 \ge 15, \\ 3x_2 + x_3 - 2x_4 = 14, \\ x_1, x_3, x_4 \ge 0, \quad x_2 \in \mathbb{R}. \end{cases}$$

Set $x_2 = p - q$ with $p, q \ge 0$, add slack s to the first constraint and subtract surplus t from the second. For $y = (x_1, p, q, x_3, x_4, s, t)^T \ge 0$,

$$\max \begin{pmatrix} -2 & 1 & -1 & -5 & 2 & 0 & 0 \end{pmatrix} y, \quad \begin{pmatrix} 1 & 1 & -1 & 2 & 5 & 1 & 0 \\ 2 & -1 & 1 & 0 & -1 & 0 & -1 \\ 0 & 3 & -3 & 1 & -2 & 0 & 0 \end{pmatrix} y = \begin{pmatrix} 20 \\ 15 \\ 14 \end{pmatrix}.$$

For a feasible original point choose $p = \max(x_2, 0)$, $q = \max(-x_2, 0)$, $s = 20 - x_1 - x_2 - 2x_3 - 5x_4$, and $t = 2x_1 - x_2 - x_4 - 15$. Conversely, any feasible y recovers a feasible original point by $x_2 = p - q$. The transformed objective is the negative of the original one.

5.2 Linear equations alone do not decide LP feasibility

The system $Ax = b$ is consistent exactly when $\operatorname{rank}(A) = \operatorname{rank}([A \mid b])$. If consistent with rank r, its solution space has $n - r$ free dimensions: uniqueness requires $r = n$. However, an LP also requires $x \ge 0$. For example, $x = -1$ is a consistent equation with no nonnegative solution. A rank test is therefore necessary but insufficient for standard-form feasibility. Equivalently, writing a_j for the columns of A, feasibility means $b = \sum_j x_j a_j$ with $x_j \ge 0$: b belongs to their conical hull.

6 Geometry of feasible sets

Definition 6.1 (Convexity)**.** A set P is convex if $\lambda u + (1 - \lambda)v \in P$ for any $u, v \in P$ and $0 \le \lambda \le 1$.

For $a \ne 0$, $a^T x = b$ defines a hyperplane, with closed halfspaces $a^T x \le b$ and $a^T x \ge b$. A *polyhedron* is the intersection of finitely many closed halfspaces; equalities can be represented by two inequalities. A bounded polyhedron is a *polytope*. These English terms distinguish what are called in Polish *zbiór wielościenny* and, in the bounded case, *wielościan*.

Proposition 6.2. *An LP feasible set is a closed convex polyhedron.*

Proof. Each linear inequality defines a closed halfspace. If $a^T u \le b$ and $a^T v \le b$, then $a^T(\lambda u + (1 - \lambda)v) \le b$. Intersections preserve convexity and closedness, including paired inequalities for equalities and halfspaces for sign restrictions. □

Definition 6.3 (Extreme point)**.** A point $v \in P$ is an extreme point (vertex) if it cannot lie strictly between two distinct points of P. Equivalently, for every $u, w \in P$ and every $0 < \lambda < 1$,

$$v = \lambda u + (1 - \lambda)w \quad \Longrightarrow \quad u = w = v.$$

The strict restriction on λ matters: taking $\lambda = 0$ or 1 cannot force both points to equal v. A point in the middle of a feasible edge is not a vertex even if it is optimal.

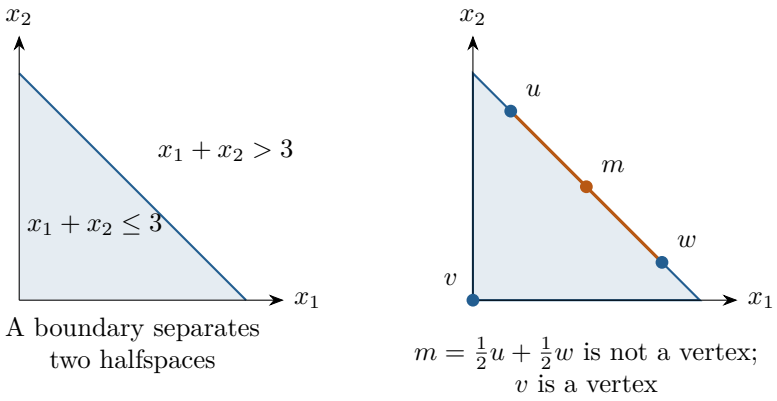


Figure 7: In two dimensions a hyperplane is a line. Intersecting its closed halfspace with $x \ge 0$ gives the shaded triangle. Convexity keeps the orange segment feasible, while the midpoint fails the extreme-point test.

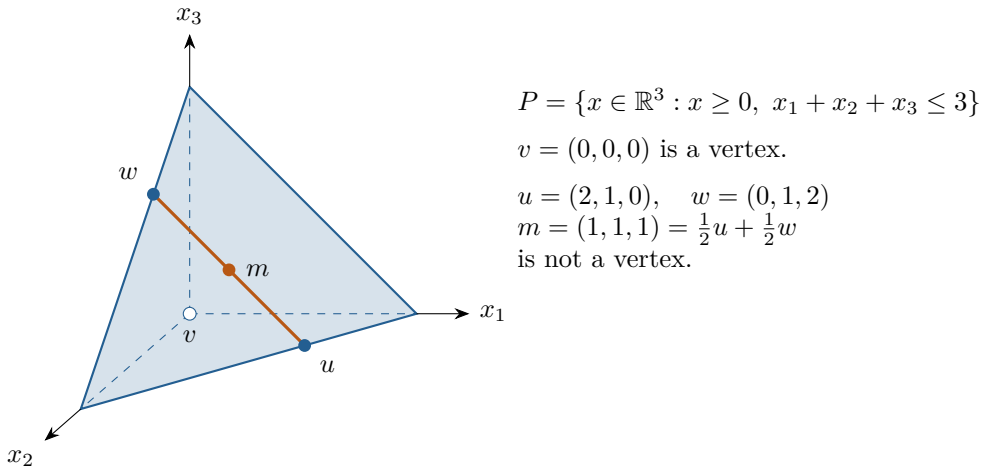


Figure 8: The same test in three dimensions. The shaded tetrahedron has a triangular face in the plane $x_1 + x_2 + x_3 = 3$. Its point m lies strictly between the distinct feasible points u and w. Viewed from the positive-coordinate side, the origin v and its three dashed edges lie behind that face.

6.1 Recession directions and extreme rays

Imagine starting at a feasible point and walking in a fixed direction. A *recession direction* lets you keep going as far as you like without leaving the feasible set, whichever feasible point you start from. Merely allowing a small step is not enough.

A nonzero vector d is a recession direction if $x + \alpha d \in P$ for every $x \in P$ and $\alpha \ge 0$; α controls how far we move. Collecting all such directions, together with zero, gives the *recession cone* $\mathrm{rec}(P)$. For a nonempty polyhedron,

$$P = \{x : Gx \le h\} \implies \mathrm{rec}(P) = \{d : Gd \le 0\},$$

because $G(x + \alpha d) = Gx + \alpha Gd$: every constraint's left side must stay constant or decrease as we move. This proves sufficiency; if a row had $G_i d > 0$, a sufficiently large step would eventually violate its bound, proving necessity. Consequently,

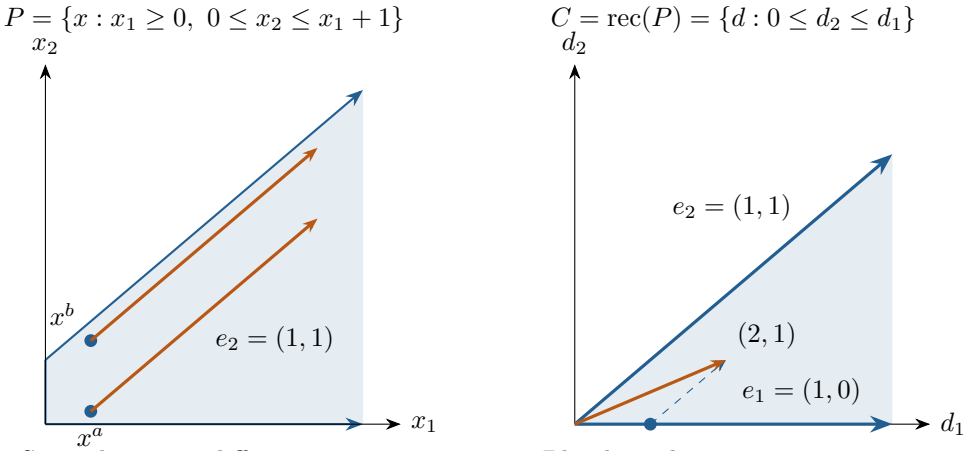
$$P = \{x : Ax = b,\ x \ge 0\} \implies \mathrm{rec}(P) = \{d : Ad = 0,\ d \ge 0\}.$$

$Ad = 0$ keeps the equalities unchanged, while $d \geq 0$ prevents any coordinate from eventually becoming negative.

A ray $\{\alpha d : \alpha \geq 0\}$ records a direction rather than a particular step length: d and $2d$ generate the same ray. A cone is *pointed* if it contains no line through zero, so it never contains both d and $-d$ for a nonzero d. An *extreme ray* is a direction that cannot be split into two allowed directions pointing elsewhere. Formally, for a pointed cone, a nonzero d generates an extreme ray if

$$d = u + v, \quad u, v \text{ in the cone} \quad \Longrightarrow \quad u = \beta d,\ v = \gamma d \text{ for some } \beta, \gamma \geq 0.$$

For example, the unbounded wedge $P = \{x : x_1 \geq 0,\ 0 \leq x_2 \leq x_1 + 1\}$ has recession cone $C = \{d : d_1 \geq 0,\ 0 \leq d_2 \leq d_1\}$. Its two boundary rays, generated by $e_1 = (1, 0)$ and $e_2 = (1, 1)$, are extreme. The direction $(2, 1)$ also allows indefinite travel, but is not extreme: $(2, 1) = e_1 + e_2$ combines two different directions in C.



Same direction, different starts; both paths continue indefinitely.

Blue boundary rays are extreme; orange direction is $e_1 + e_2$.

Figure 9: Left: a recession direction can be followed forever from any feasible starting point. Right: the recession cone contains direction vectors, not starting points. The dashed arrow is a translated copy of e_2, showing why $(2, 1)$ is not extreme. Shading continues beyond the right edge of each drawing.

A nonempty polyhedron is unbounded exactly when its recession cone has a nonzero vector. The forward implication follows from a finite polyhedral representation; the reverse follows from the feasible ray: there is no limit to how far one can travel. This concerns the size of the feasible set, not whether an objective improves along that path.

Closedness matters beyond polyhedra. The nonclosed convex set $\{(0,0)\} \cup \{(u,v) : u > 0,\ 0 < v < 1\}$ is unbounded but has no nonzero recession direction. Any ray from the origin either leaves $0 < v < 1$ eventually or stays on an excluded part of its boundary. The equivalence holds for nonempty closed convex sets, including polyhedra.

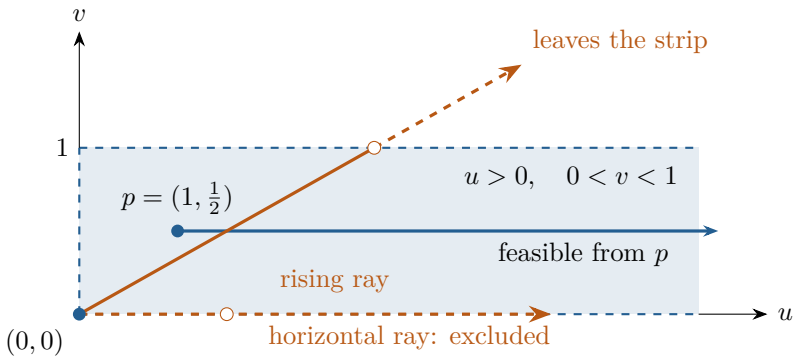


Figure 10: An unbounded convex set with no nonzero recession direction. Dashed boundaries are excluded, except for the included origin (filled dot). From the origin, a rising ray reaches the excluded top edge; a horizontal ray follows the excluded bottom edge. The blue ray stays feasible from p, but its direction fails at the origin. Shading continues to the right.

7 Existence, unboundedness, and the vertex principle

For an LP there are three objective outcomes: infeasible; feasible with objective unbounded in the improving direction; or feasible with a finite attained optimum. The last case may have one or infinitely many optimal decisions.

7.1 Unbounded feasible set versus unbounded objective

A *feasible ray* is a half-line $\{x^0 + \alpha d : \alpha \geq 0\}$ contained entirely in P: start at the feasible point x^0 and move in a fixed nonzero direction d. Increasing α takes us farther from the starting point without violating any constraint.

If x^0 is feasible in standard form and $Ad = 0$, $d \geq 0$, $c^T d > 0$, then

$$x^0 + \alpha d \in P, \qquad c^T(x^0 + \alpha d) = c^T x^0 + \alpha c^T d \xrightarrow{\alpha \to +\infty} +\infty.$$

Here x^0, d, and c stay fixed; α grows without bound. The positive number $c^T d$ is the objective gain per unit increase in α, so continued travel keeps improving a maximization objective.

Feasibility is an essential assumption: a direction alone does not cure an empty feasible set. For example, with $x_1 - x_2 = 1$, $x \geq 0$, maximizing x_1, use $x^0 = (1,0)^T$ and $d = (1,1)^T$. The objective is $1 + \alpha$.

On the same ray, $-x_1 = -1 - \alpha$ decreases as α grows, so maximizing $-x_1$ instead chooses $\alpha = 0$, giving the unique optimum $(1,0)$. This is also the solution of minimizing x_1: making a value smaller makes its negative larger. The optimal point is the same, while the optimal values have opposite signs: $\min x_1 = -\max(-x_1) = 1$. Thus an unbounded feasible set can still have a finite optimum.

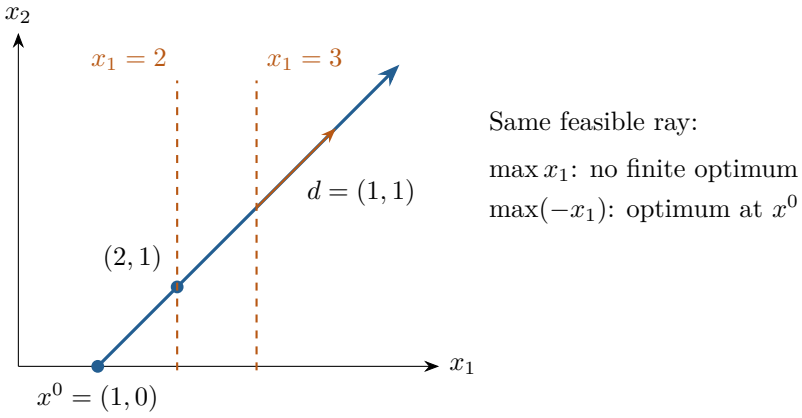


Figure 11: The feasible ray $x = x^0 + \alpha(1, 1)$, $\alpha \geq 0$, is unbounded. For maximizing x_1, the orange level lines can move indefinitely to the right. Changing the objective to $-x_1$ makes the ray's endpoint uniquely optimal.

7.2 Representation by vertices and rays

Theorem 7.1 (Polyhedral representation, stated without full proof)**.** *For a nonempty pointed polyhedron P, let $v^1, \ldots, v^k$ be its vertices and $d^1, \ldots, d^\ell$ representatives of its extreme recession rays. Then*

$$x \in P \quad \Longleftrightarrow \quad x = \sum_{i=1}^{k} \lambda_i v^i + \sum_{j=1}^{\ell} \mu_j d^j, \qquad \lambda_i, \mu_j \geq 0, \quad \sum_i \lambda_i = 1.$$

For bounded P, the ray sum is absent.

Geometrically, every feasible point can be reached in two steps. First choose a weighted average of the vertices (the corners): the λ_i are shares that add to 1. Then move along the extreme recession directions (the directions of escape): the μ_j are nonnegative step sizes and need not add to 1. Conversely, every point constructed this way remains feasible. If P is bounded, averaging its vertices alone fills the whole polyhedron; there are no directions of escape to add.

Standard-form feasible sets are pointed: a line contained in $x \geq 0$ would force its direction to be zero. Thus the theorem applies when they are nonempty. Polyhedra containing lines require an additional lineality term.

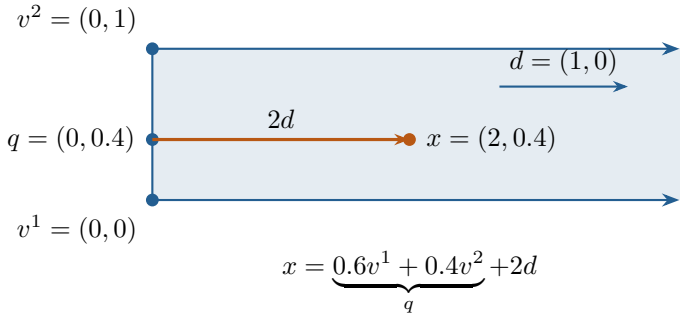


Figure 12: Vertex–ray representation for $P = \{x : x_1 \geq 0,\ 0 \leq x_2 \leq 1\}$. First choose a convex combination on the left edge, then add a nonnegative multiple of the horizontal extreme recession direction.

Theorem 7.2 (Optimal vertex for the lecture's standard form)**.** *Let $P = \{x : Ax = b,\ x \geq 0\} \neq \varnothing$. If $c^T x$ is bounded above on P, its maximum is attained at a vertex of P.*

Proof using Theorem 7.1. If any $c^T d^j > 0$, its feasible ray makes the objective unbounded. Hence all ray contributions are nonpositive, and every feasible x satisfies

$$c^T x = \sum_i \lambda_i c^T v^i + \sum_j \mu_j c^T d^j \le \sum_i \lambda_i c^T v^i \le \max_i c^T v^i.$$

A vertex attaining that finite maximum is feasible and optimal. □

This is an *existence* statement: not every optimum must be a vertex. It also cannot be applied without qualification to arbitrary feasible sets. For example, maximize $-x_1$ on $\{x \in \mathbb{R}^2 : x_1 \ge 0\}$. The optimum is zero on an entire line, and that halfspace has no vertices.

Proposition 7.3 (Multiple optima)**.** *Every convex combination of optimal LP solutions is optimal.*

Proof. For optimal u, v with common value z^*, feasibility follows from convexity and $c^T(\lambda u + (1-\lambda)v) = \lambda z^* + (1-\lambda)z^* = z^*$. □

In $x_1 + x_2 \le 1$, $x \ge 0$, maximizing $x_1 + 2x_2$ gives $(0, 1)$ uniquely; maximizing $2x_1 + x_2$ gives $(1, 0)$ uniquely. Maximizing $2x_1 + 2x_2$ makes the whole edge $x_1 + x_2 = 1$ optimal.

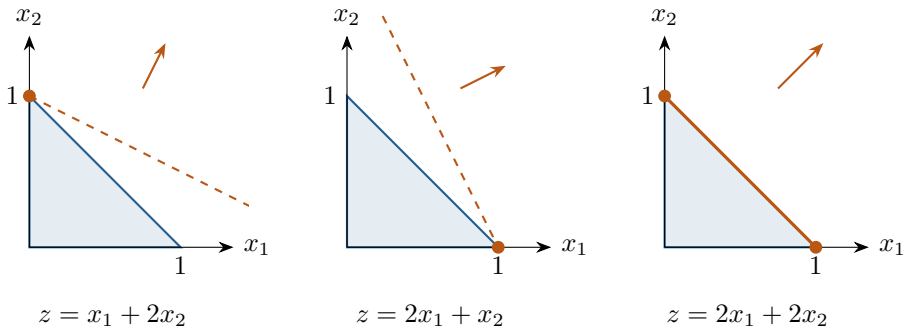


Figure 13: One feasible triangle, three objectives. The first two have different unique optimal vertices. In the third, the supporting objective line coincides with the whole sloping edge, so every point on that edge is optimal. Orange arrows indicate objective increase.

The representation above also allows zero-objective recession directions: optimal decisions can lie on unbounded optimal rays, not just between vertices.

8 Bases and basic feasible solutions

Assume $A \in \mathbb{R}^{m \times n}$ has full row rank $m \le n$. Choose m independent columns indexed by B; their square matrix A_B is nonsingular. With N the remaining indices, define

$$x_N = 0, \qquad x_B = A_B^{-1} b.$$

This is a *basic solution.* It is *basic feasible* (BFS) only if $x_B \ge 0$. Compute it by solving $A_B x_B = b$ rather than forming an inverse.

Theorem 8.1. *A feasible point of $\{x : Ax = b, x \ge 0\}$ is a vertex exactly when it is a BFS.*

Proof. For a BFS $x = \lambda u + (1-\lambda)v$, $0 < \lambda < 1$, nonnegativity and $x_N = 0$ force $u_N = v_N = 0$. Then $A_B u_B = A_B v_B = b$, so nonsingularity yields $u = v = x$.

Conversely, let $S = \{j : x_j > 0\}$ for a vertex x. If the columns A_S were dependent, some nonzero d supported on S would satisfy $Ad = 0$. For small $\varepsilon > 0$, both $x + \varepsilon d$ and $x - \varepsilon d$ would remain nonnegative and feasible, contradicting extremality. Thus A_S is independent and can be extended to a basis B of m columns. The other variables are zero, so x is its basic feasible solution. □

A BFS has at most m positive coordinates. That count alone is insufficient: the corresponding columns must be independent. For example,

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad b = (2,0)^T, \quad x = (1,1,0)^T$$

has only $m = 2$ positive coordinates but is the midpoint of two feasible points, $(2,0,0)^T$ and $(0,2,0)^T$, not a vertex. A BFS is *degenerate* when at least one of its m basic coordinates is zero; different bases can then represent the same vertex.

8.1 Production example in standard form

Add three slacks to Equation 1. The augmented vector is $y = (x_1, x_2, s_1, s_2, s_3)^T$, with

$$\begin{pmatrix} 2 & 1 & 1 & 0 & 0 \\ 3 & 3 & 0 & 1 & 0 \\ 1.5 & 0 & 0 & 0 & 1 \end{pmatrix} y = \begin{pmatrix} 10 \\ 24 \\ 6 \end{pmatrix}, \qquad y \geq 0.$$

At the origin in production coordinates, $B = \{3,4,5\}$ is the slack basis. At the optimum, $B = \{1,2,5\}$ yields $(x_1, x_2, s_3) = (2,6,3)$ and $s_1 = s_2 = 0$. The determinant of this basis matrix is $3 \neq 0$. The graphical intuition of simplex is to move between adjacent basic feasible solutions toward improving objective values. Pivot rules, degeneracy handling, and termination arguments require later algorithmic material; Lecture 2 does not yet supply a complete simplex implementation.

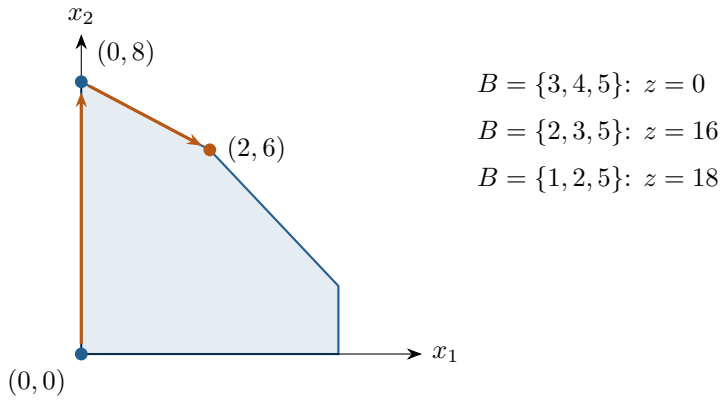


Figure 14: An improving path through adjacent production vertices and their bases in coordinates $(x_1, x_2, s_1, s_2, s_3)$. At $(0,8)$ the basic coordinates are $(x_2, s_1, s_3) = (8,2,6)$ and the second resource is tight ($s_2 = 0$). The path illustrates basis changes without specifying a pivot rule.

9 MATLAB checks

The shared script is `matlab/exercises/lecture_02_checks.m`. It enumerates production vertices, checks resource slacks and the optimal basis, verifies the 5.2 m cutting certificate, and enumerates the explicitly assumed 2 m cabinet problem. It uses base MATLAB operations without Optimization Toolbox. The inputs match the hand derivations. Assertions use absolute tolerance 10^{-10} for these small examples; this is not a universal numerical tolerance.

10 Common mistakes and revision questions

- Do not confuse unbounded feasible sets with objectives unbounded above.

- Do not call every feasible intersection a vertex; discard intersections violating other constraints and check independent active constraints.
- Do not confuse slack and surplus signs or omit nonnegativity.
- Do not replace integer board counts by arbitrary fractional values without calling the resulting problem an LP relaxation.
- Do not replace exact demand by minimum demand without considering surplus.
- A feasible basic solution needs nonnegative basic coordinates; a nonsingular basis alone does not guarantee feasibility.

For revision: derive the production certificate without the figure; transform a free variable and a $\geq$ constraint; construct an infeasible LP whose equations are consistent; and explain why an optimal edge does not contradict the vertex theorem.

11 Sources and editorial remarks

Primary source: `s4_w2.pdf`, PDF pages 1–11, printed slides 1–44.
Source date: Not established; stored in the folder labelled 2020.
Current course alignment: `ProgMat_w1.pdf`, pp. 8–9, 14–15, 19, and 27–30, confirms topical overlap for resource models, diet, transportation, regression, standardization and vertex motivation. The current transport example uses exact demand and prohibited routes; it is developed in `02a-linear-modeling-patterns.tex`. Historical board-cutting material and the full geometric proofs are additional study material.
Editorial scope: Source models and results are reorganized in English. Hand derivations, proofs, cautions, and MATLAB checks are added study material.

The two-food diet and three-observation regression examples in Section 3 are added hand-checkable illustrations, with their own bound certificates. The actual new companion is modeling-patterns MATLAB script at `matlab/exercises/modeling_patterns_checks.m`, using the same data and absolute tolerance 10^{-10}. Their exact calculations were independently checked. The user supplied successful output on 2026-10-11 and confirmed MATLAB 26.2.0.3386108 (R2026b); all modeling-pattern assertions passed. That run used the earlier diet coefficients. The revised diet result 28/5 was checked in exact rational arithmetic, and the updated script passed static analysis. On 2026-10-19, the agent executed the revised script on R2026b; all assertions passed, including the revised diet certificate. The actual command and output are recorded in `docs/matlab-verification.md`. This companion covers the diet and regression examples, while the historical script's execution below concerns its own examples.

PDF pages	Printed slides	Content
1–2	1–8	Sections 1–2
3–5	9–20	Sections 3–5
6	21–24	Section 5
7–8	25–32	Sections 6–7
9–10	33–40	Sections 6–7
11	41–44	Section 8

The main source is `s4_w2.pdf`; added proofs and worked solutions are editorial explanations based on its models. The regression formula and cabinet sketch were inspected visually because PDF text extraction loses their notation. Polish vocabulary is maintained in `docs/glossary.md`. The Polish cabinet prompt in Section 4 is from slide 19; the English translation and idealized cutting solution are additions. Figures 1–14 are redrawn explanations; the transportation network, regression band, and strip representation use illustrative configurations.

Supplementary cross-check: MIT OCW, 15.081J/6.251J, Lecture 10 (Fall 2009), PDF pp. 4 and 6, sections on recession cones, unbounded LPs, and the resolution theorem. Used to check hypotheses of the vertex–ray representation; this supplement is not current course coverage. The source inventory records its external storage and download status.

Numerical verification. Section 4.3 and Figure 6 are original added examples; they do not change the supplied cutting data or their MATLAB companion. The exact bounds 29/4 and 7 follow from the displayed weighted inequalities. For further study of relaxations, cuts and branching, see MIT OCW, *Discrete optimization II*, Sections 2–3. Current-course alignment of this extension is unknown.

Verified in MATLAB R2026b on 2026-10-04, using the user's normal terminal. All assertions passed. The reported output matches the hand derivations: production $(2, 6)$ with value 18 PLN; 5.2 m cutting $(200, 0, 400)$ with offcut 140 m; and 2 m cabinet count 11 with pattern counts $(10, 1, 0)$. The script also asserts the production slacks $(0, 0, 3)$ kg. No Optimization Toolbox solver is invoked. The supplied terminal output is recorded in `docs/matlab-verification.md`.

Terminology and hypotheses. In Section 7, the pointed vertex–ray result is a polyhedral representation (or resolution) theorem. The historical slides call it Carathéodory's theorem; the convex-hull theorem of that name instead reduces a convex combination to at most $n+1$ points. The MIT supplement was used to check pointedness. In Section 6, the recession equivalence is stated with closedness; the historical version omits that hypothesis. In Section 2, selling-price data justify revenue, whereas the slides also use the word profit.